



15th International Conference on Greenhouse Gas Control Technologies, GHGT-15

15th 18th March 2021 Abu Dhabi, UAE

Synthesis and characterization of Ca-based layered double hydroxides as solid sorbents for carbon capture

Maria C. Curria^{a,b}, Claire E. White^{a,b*}

^a Department of Civil and Environmental Engineering, Princeton University, Princeton 08544, USA

^b Andlinger Center for Energy and the Environment, Princeton University, Princeton, 08544, USA

Abstract

The development of new sorbents for carbon capture that are (i) widely available, (ii) low cost, and (iii) require less energy for rejuvenation compared with current sorbent technologies is an area of significant interest to industries with point-source CO₂ emissions and those pursuing direct air capture (DAC). Here, we synthesized high purity Ca-based layered double hydroxides (LDHs) and performed a preliminary evaluation of their low-temperature carbon capture potential at high relative humidity (85 %). In particular, CaFe- and CaAl-LDHs with chloride as the interlayer anion were obtained through addition of a trivalent cation (Fe³⁺ or Al³⁺) to a Ca(OH)₂ suspension in an inert environment. Characterization of the reaction products was carried out using scanning electron microscopy (SEM), powder X-ray diffraction (XRD) and attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy, where preliminary results showed successful synthesis of high-purity Ca-based LDHs. The synthesized samples were carbonated at 30 °C for 1 hr in a controlled environment with 85 % RH and 1 % CO₂. The carbonated samples were characterized using XRD and ATR-FTIR analysis, and subsequently subjected to thermal decomposition via TGA where the evolved gases were studied concurrently using FTIR analysis. Our results show that the chemical tailoring of Ca(OH)₂ may not only reduce its regeneration energy but also lead to an improved adsorption capacity.

Keywords: layered double hydroxides, portlandite, low-temperature carbon capture, solid sorbents

1. Introduction

Currently, the capture stage of carbon capture and sequestration (CCS) operations accounts for ~70 – 90 % of the total cost, and therefore the identification and implementation of new sorbents that can significantly reduce the overall cost of CCS is an active area of research [1]. One of the technologies currently being used for carbon capture is calcium looping, which uses CaO (or Ca(OH)₂ if the loop includes a hydration step) as a sorbent to separate CO₂ at high temperature (usually above 500 °C) by means of a chemisorption mechanism that forms CaCO₃ [2]. Hydrating the rejuvenated CaO to form Ca(OH)₂ (called portlandite in its crystalline form) reverts the sintering process suffered during rejuvenation that is known to decrease the cyclability of the sorbent [2]. Although CaO and Ca(OH)₂ are widely available low-cost materials, the CaCO₃ formed upon chemisorption of the captured CO₂ requires extremely high temperatures (usually above 900 °C [3]) to desorb the CO₂ and rejuvenate the sorbent. In other words, there is a very high energy penalty associated with the use of these sorbents for carbon capture, particularly when trying to separate the CO₂ from low-temperature gas streams, as it is the case for DAC and many CO₂-intensive industries.

* Corresponding author. Tel.: +1-609-258-6263, E-mail address: whitece@princeton.edu

Layered double hydroxides (LDHs) are bimetallic hydroxides where the difference in oxidation states of the two metal cations in the hydroxide layer creates a positive charge that is balanced by interlayer anions. The general composition of LDHs is usually expressed as $[M^{2+}_{1-x} M^{3+}_x (OH)_2]^{x+} [A^{n-}_{x/n} \cdot mH_2O]^{x-}$, where $[M^{2+}_{1-x} M^{3+}_x (OH)_2]^{x+}$ is the bimetallic hydroxide layer consisting of the M^{2+} and M^{3+} cations and $[A^{n-}_{x/n} \cdot mH_2O]^{x-}$ is the interlayer composition consisting of the A^{n-} anion and m -interlayer water molecules [4–10]. Ca-based LDHs, including hydrocalumite ($Ca_2Al(OH)_6[Cl_{1-x}(OH)_x] \cdot 2H_2O$), have recently gained attention as potential sorbents for carbon capture applications [11–13]. Ca-based LDHs possess a similar molecular structure to Mg-based LDHs (for example, hydrotalcite), which have been extensively studied for carbon capture applications, however, the enhanced surface basicity of Ca-based LDHs compared with Mg-based LDHs due to the higher electronegativity of the Ca(II) with respect to Mg(II) [11,12,14] may lead to an increased CO_2 adsorption strength.

Here we have investigated the low-temperature chemisorption of CO_2 on crystalline Ca-based LDHs ($Ca_2Fe(OH)_6Cl \cdot 2H_2O$ and $Ca_2Al(OH)_6Cl \cdot 2H_2O$) and portlandite. The high-purity of the as-synthesized samples was confirmed using powder X-ray diffraction (XRD) and attenuated total reflectance Fourier-transform infrared spectroscopy (ATR-FTIR). ATR-FTIR and XRD were also used to characterize the carbonated samples, where the formation of carbonated phases was detected in the Ca-based LDHs and portlandite. The thermal decomposition of the carbonated samples was studied using thermogravimetric analysis (TGA) coupled with FTIR analysis of the evolved gases, showing that the Ca-based LDHs offer enhanced adsorption capacity and lower regeneration temperature compared with portlandite. These preliminary results indicate that it may be possible to chemically tailor Ca-based LDHs for optimal performance according to specific carbon capture requirements.

2. Materials and methods

2.1. Synthesis

$Ca_2Fe(OH)_6Cl \cdot 2H_2O$ and $Ca_2Al(OH)_6Cl \cdot 2H_2O$ samples (denoted as CaFeCl-LDH and CaAlCl-LDH, respectively) were synthesized by a modified coprecipitation method as explained in ref. [15]. The $Ca(OH)_2$ suspension (CaO ReagentPlus 99.9 % trace metals basis – Sigma, in 100 ml of MilliQ) was titrated with a MCl_3 solution (for $M = Fe$, $FeCl_3 \cdot 6H_2O$ Reag. Ph. Eur., $\geq 99\%$ - Sigma Aldrich; for $M = Al$, $AlCl_3$ ReagentPlus 99 % - Sigma Aldrich) at a rate of ≈ 1 ml/min. Recrystallized portlandite was precipitated from a saturated $Ca(OH)_2$ suspension for use as a reference material. All precipitated solids were separated from the supernatant by filtration and dried at room temperature for 24 hrs. Once dried, the samples were ground using a mortar and pestle and stored in air-tight containers. This synthesis procedure was conducted inside an N_2 glovebox to prevent early carbonation. Controlled carbonation of the samples was carried out in a carbonation chamber for 1 hr at a temperature of $30^\circ C$, 1 % CO_2 concentration and 85 % relative humidity.

2.2. Characterization

XRD patterns were obtained using a Bruker D8 Advance X-ray diffractometer with Ag-radiation ($\lambda_{K\alpha 1} = 0.55941$ Å, $\lambda_{K\alpha 2} = 0.56380$ Å). The powder samples were loaded in 1 mm outer diameter polyimide capillaries and sealed at both ends with quick set epoxy. Scans were performed over a 2θ range of 1.5° to 30° with a step size of 0.01° and a dwell time of 2 s per step. Phase identification was carried out using the Bruker EVA software.

The vibrational modes of the samples were probed using ATR-FTIR with a Perkin Elmer Frontier MIR instrument and a Frontier UATR accessory. For each sample, 32 scans were collected from 510 to 4000 cm^{-1} at a scan speed of 0.1 cm/s with a resolution of 1 cm^{-1} . The instrument and UATR accessory were purged using N_2 to avoid contributions from atmospheric H_2O and CO_2 .

The surface morphology of the samples was studied using scanning electron microscopy (SEM). An XL30 FEG-SEM instrument was employed, where powder samples were mounted on SEM holders using carbon tape. Images were obtained in an ultra-high vacuum environment with a 5 keV beam and a working distance of 10 mm.

The thermal decomposition of carbonated samples was studied using a Perkin Elmer Pyris 1 TGA instrument. The TGA temperature program consisted of a 30 min isothermal step at 50 °C followed by ramping to 900 °C at a rate of 10 °C/min. The evolved gases from the TGA sample were studied using a Perkin Elmer Frontier MIR instrument, a TL 8000 transfer line and a gas IR cell under a N₂ purge. The transfer line was set to a flow rate of 20 mL/min and a temperature of 300 °C. The IR cell was set at a temperature of 250 °C. The FTIR spectra of the evolved gases were measured every 30 secs over a wavelength range of 550 to 4000 cm⁻¹, a scan speed of 1cm/s and a resolution of 1 cm⁻¹.

3. Results and discussion

The XRD patterns of the CaFeCl-LDH and CaAlCl-LDH samples before and after carbonation are presented in Fig 1, where it is clear that calcite is the main carbonation product in both systems. There is also a secondary amorphous phase (or phases) in these carbonated samples, as seen by the presence of weak diffuse scattering at $2\theta \approx 13^\circ$ for carbonated CaFeCl-LDH and CaAlCl-LDH, and at $2\theta \approx 7^\circ$ for carbonated CaAlCl-LDH. Moreover, the carbonated CaAlCl-LDH sample contains a Bragg peak at $2\theta \approx 7.4^\circ$ (d -spacing of 4.3 Å), that could not be assigned to any of the expected phases.

The ATR-FTIR spectra of carbonated CaFeCl-LDH, CaAlCl-LDH and portlandite shown in Fig. 2c and 2d confirm that the carbonation experiment leads to chemisorption of CO₂ and the formation of calcite in all samples (vibrational modes at 712, 872 and 1401 cm⁻¹) [16]. Interestingly, the stretching modes of M³⁺-O at 741 cm⁻¹ for CaFeCl-LDH and 785 cm⁻¹ for CaAlCl-LDH [17] do not appear in the spectra of carbonated CaFeCl-LDH and CaAlCl-LDH, which indicates a change in their chemical environment. The additional vibrational modes of carbonated CaAlCl-LDH at 745, 916, 968 and 1022cm⁻¹ resemble that of gibbsite, a polymorph of Al(OH)₃ [18], however, the XRD pattern presented in Fig 1b does not show any Bragg reflections associated with this crystalline phase. Further investigation needs to be performed to characterize the change in the chemical environment of the M³⁺ upon carbonation.

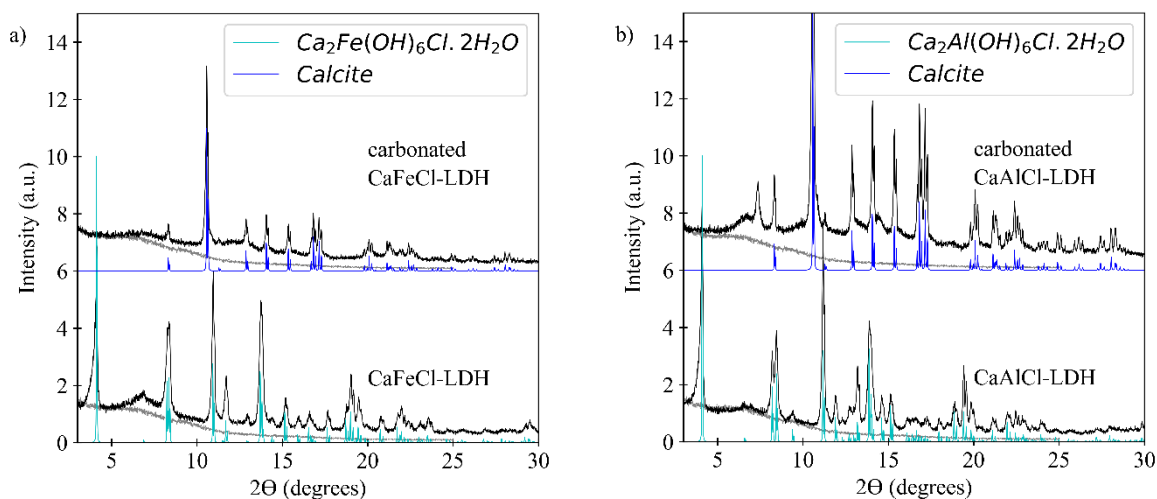


Fig 1. XRD patterns of (a) CaFeCl-LDH and (b) CaAlCl-LDH, before and after carbonation. Carbonation was carried out for 1 hr at 30 °C in a controlled environment with 85 % RH and 1 % CO₂.

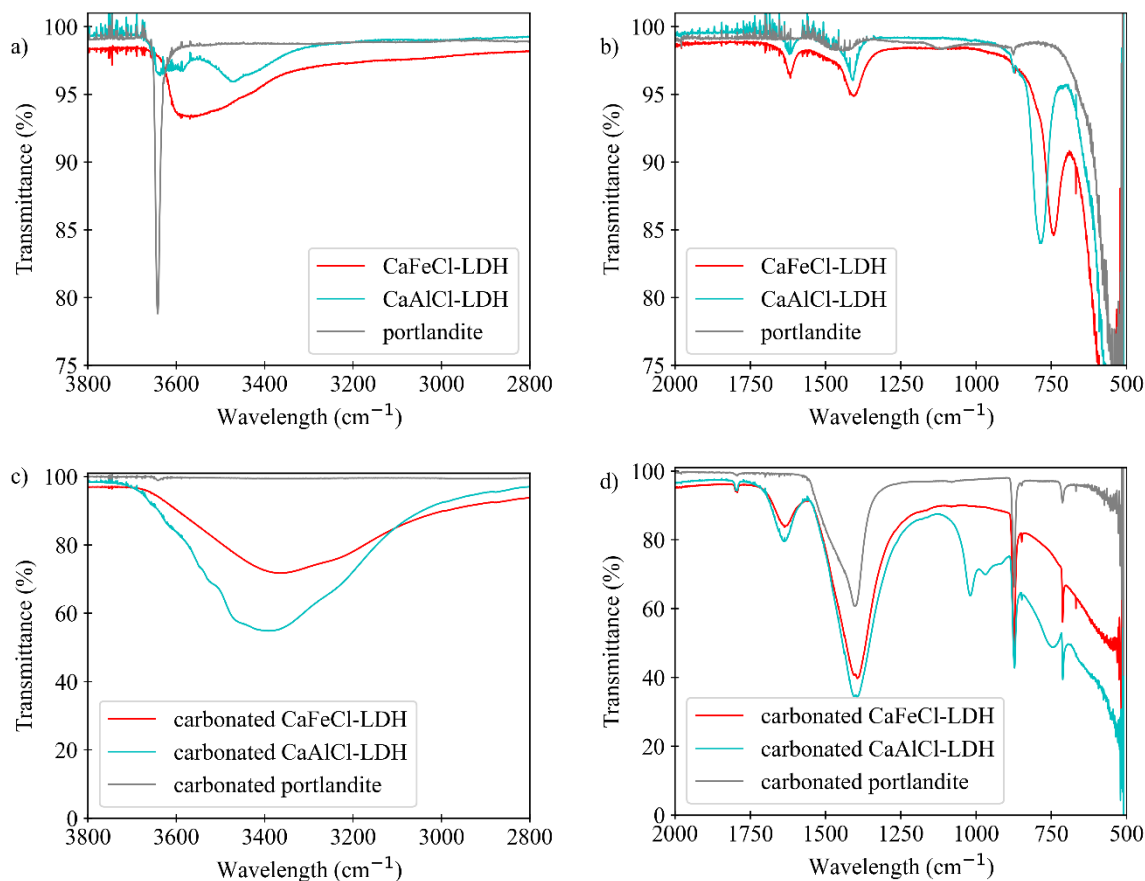


Fig. 2. ATR-FTIR spectra of CaFeCl-LDH, CaAlCl-LDH and recrystallized portlandite (a, b) before and (c, d) after carbonation.

TGA coupled with FTIR spectroscopy of the evolved gases has been used to provide further insight on the carbonate phase(s) and other chemical environments present in the carbonated samples. At temperatures below ~ 500 °C the weight loss regions of carbonated CaFeCl-LDH, CaAlCl-LDH and portlandite are attributed to the loss of water molecules and hydroxyl units since the corresponding gas-based FTIR spectra observed for these decomposition events resemble the spectrum shown in Fig. 3b [19]. Specifically, the weight loss associated with carbonated CaFeCl-LDH and CaAlCl-LDH at a temperature of ~ 100 °C is due to the loss of adsorbed water and interlayer water molecules [20] found in any residual LDH, while at higher temperatures (e.g., ~ 270 °C and above) the hydroxyl units will be lost from the residual LDH and potentially the decomposition of any secondary hydroxylated phases that may have formed during the carbonation process. For carbonated portlandite the weight loss at ~ 470 °C is associated with dehydroxylation of neat portlandite that remains in the sample [21].

The TGA data presented in Fig. 3a show two distinct decomposition events for both the carbonated CaFeCl-LDH and CaAlCl-LDH samples at temperatures above ~ 520 °C while carbonated portlandite undergoes a single decomposition event at ~ 740 °C. The corresponding FTIR spectra of the evolved gases were found to resemble that of CO₂ (see Fig. 3c), with intense vibrational modes at ~ 2350 cm⁻¹ along with another distinct vibrational mode at 668 cm⁻¹ [19]. The CO₂ desorption at high temperature (> 700 °C) in all three samples (carbonated CaFeCl-LDH, CaAlCl-LDH and portlandite) is attributed to the decomposition of calcite. However, it is unclear what is responsible for the lower temperature decomposition event in carbonated CaFeCl-LDH and CaAlCl-LDH at ~ 620 °C. One possible

explanation may involve the amorphous phase(s) seen in the XRD data and therefore additional experiments are required to determine its chemical composition and structure. It is also worth noting that the relative amount of these two carbonated phases, as determined from the derivative TGA curves in Fig. 3a, is fairly evenly weighted in the carbonated CaFeCl-LDH sample while there is more calcite in the carbonated CaAlCl-LDH sample.

Interestingly, both CaFeCl-LDH and CaAlCl-LDH show higher extent of carbonation compared with portlandite. This is evident by the percentage weight loss between 510 and 800 °C in Fig. 3a, where carbonated portlandite lost 12 % of its initial weight while carbonated CaFeCl-LDH and CaAlCl-LDH lost 16 % and 18 %, respectively. The more extensive carbonation of the Ca-based LDHs compared with portlandite may be attributed to their exfoliated morphology that provides more adsorption sites than the relatively smooth-surfaced hexagonal particles of portlandite, as seen in the SEM images presented in Fig. 4.

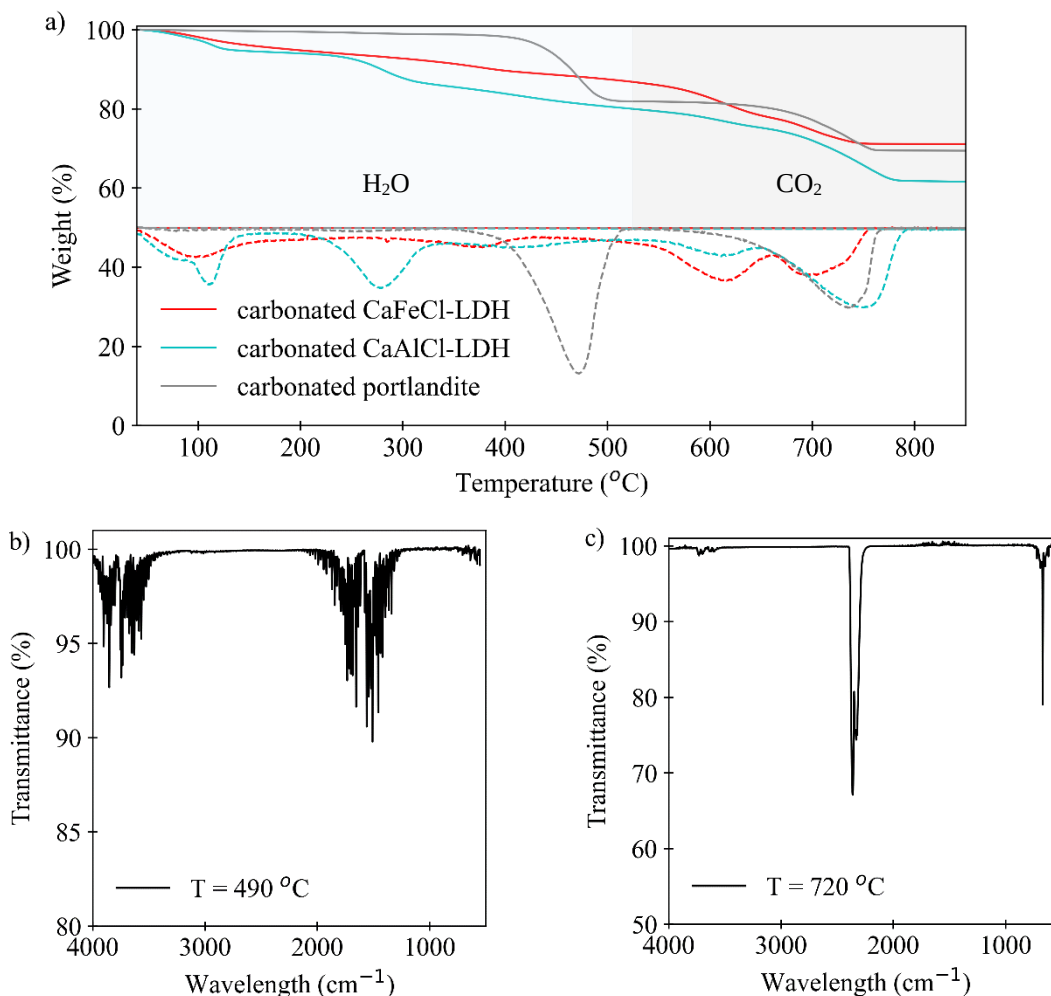


Fig. 3. (a) Thermogravimetric analysis (solid) and differential thermal analysis (dashed) curves of carbonated CaFeCl-LDH, CaAlCl-LDH and portlandite. (b) and (c) Gas-FTIR spectra collected during the thermal decomposition of carbonated portlandite at 490 °C and 720 °C, respectively.

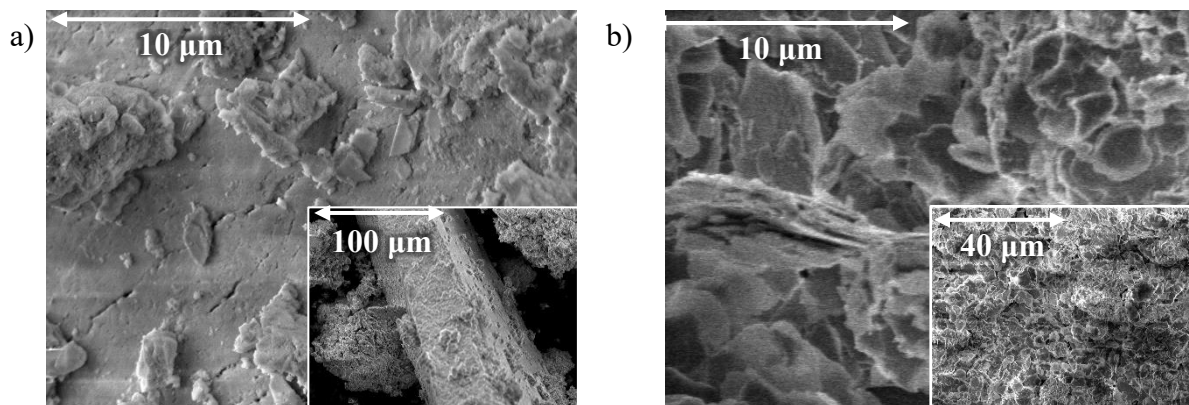


Fig. 4. Scanning electron microscope images of (a) recrystallized portlandite and (b) CaFeCl-LDH.

Conclusions

Crystalline $\text{Ca}_2\text{Fe}(\text{OH})_6\text{Cl}\cdot 2\text{H}_2\text{O}$ and $\text{Ca}_2\text{Al}(\text{OH})_6\text{Cl}\cdot 2\text{H}_2\text{O}$ were synthesized using a modified coprecipitation method via the mixing of MCl_3 (M denotes trivalent metal) with a $\text{Ca}(\text{OH})_2$ suspension. The CO_2 chemisorption performance of these Ca-based LDHs was compared to that of portlandite, where XRD, ATR-FTIR and TGA coupled with FTIR analysis of the evolved gases indicated that an additional carbonate phase formed in the carbonated LDH phases with a decomposition temperature $\sim 100^\circ\text{C}$ lower than that of calcite. Moreover, both Ca-based LDHs showed a higher CO_2 adsorption capacity than portlandite, likely due to their enhanced surface area.

Acknowledgments

The authors acknowledge the use of the Imaging and Analysis Center at Princeton University, which is partially supported by the Princeton Center for Complex Materials, a National Science Foundation (NSF)-MRSEC program (DMR-1420541). We also acknowledge the financial support received from the Andlinger Center for Energy and the Environment, and travel support from the Peter B. Lewis Fund for Student Innovation in Energy and the Environment and the School of Engineering and Applied Science at Princeton University.

References

- [1] Folger P. Carbon Capture: A Technology Assessment. Technol Issues Carbon Capture 2010.
- [2] Phalak N, Wang W, Fan LS. $\text{Ca}(\text{OH})_2$ -Based Calcium Looping Process Development at The Ohio State University. Chem Eng Technol 2013; 36:1451–1459.
- [3] Florin N, Fennell P. Synthetic CaO-based Sorbent for CO_2 Capture. Energy Procedia. 2011.
- [4] Miyata S. The Syntheses of Hydrotalcite-Like Compounds and Their Structures and Physico-Chemical Properties I: The Systems $\text{Mg}^{2+}\text{-Al}^{3+}\text{-NO}_3^-$, $\text{Mg}^{2+}\text{-Al}^{3+}\text{-Cl}^-$, $\text{Mg}^{2+}\text{-Al}^{3+}\text{-ClO}_4^-$, $\text{Ni}^{2+}\text{-Al}^{3+}\text{-Cl}^-$ and $\text{Zn}^{2+}\text{-Al}^{3+}\text{-Cl}^-$. Clays Clay Miner 1975; 23:369–375.
- [4] Miyata S. Synthesis of Hydrotalcite-Like Compounds and their Physico-Chemical Properties—The Systems $\text{Mg}^{2+}\text{-Al}^{3+}\text{-SO}_4^{2-}$ and $\text{Mg}^{2+}\text{-Al}^{3+}\text{-CrO}_4^{2-}$. Clays Clay Miner 1977; 25:14–18.
- [5] Miyata S. Adsorption of N_2 , O_2 , CO_2 and H_2 on Hydrotalcite-Like System: $\text{Mg}^{2+}\text{-Al}^{3+}\text{-(Fe(CN)}_6\text{)}^{4-}$. Clays Clay Miner 1978; 26:441–447.
- [7] Long X, Wang Z, Xiao S, An Y, Yang S. Transition metal based layered double hydroxides tailored for energy conversion and storage. Mater Today 2016; 19:213–226.
- [8] Yang Z zhu, Wei J jing, Zeng G ming, Zhang H qing, Tan X fei, Ma C, Li X cheng, Li Z hao, Zhang C. A review on strategies to LDH-based materials to improve adsorption capacity and photoreduction efficiency for CO_2 . Coord Chem Rev 2019; 386:154–182.
- [9] Gevers BR, Labuschagné FJWJ. Green Synthesis of Hydrocalumite (CaAl-OH-LDH) from $\text{Ca}(\text{OH})_2$ and $\text{Al}(\text{OH})_3$ and the Parameters That Influence Its Formation and Speciation. Crystals 2020; 10:672.
- [10] Labuschagné FJWJ, Wiid A, Venter HP, Gevers BR, Leuteritz A. Green synthesis of hydrotalcite from untreated magnesium oxide and

- aluminum hydroxide. *Green Chem Lett Rev* 2018; 11:18–28.
- [11] Narayanappa AN, Kamath PV. Interaction of Pristine Hydrocalumite-Like Layered Double Hydroxides with Carbon Dioxide. *ACS Omega* 2019; 4:3198–3204.
- [12] Rossi TM, Campos JC, Souza MMVM. An evaluation of calcined hydrocalumite as carbon dioxide adsorbent using thermogravimetric analysis. *Appl Clay Sci* 2019; 182.
- [13] Manohara G V, Maroto-Valer MM, Garcia S. The effect of the layer-interlayer chemistry of LDHs on developing high temperature carbon capture materials. *Dalt Trans* 2020; 49:923–931.
- [14] López-Salinas E, Serrano MEL, Jácome MAC, Secora IS. Characterization of synthetic hydrocalumite-type $[\text{Ca}_2\text{Al}(\text{OH})_6]\text{NO}_3 \cdot m\text{H}_2\text{O}$: Effect of the calcination temperature. *J Porous Mater* 1996; 2:291–297.
- [15] Kim JH, Park M, Imran A, Choi M-C, Kim KS, Komarneni S. Post-isomorphic substitution of trivalent metal cations for Ca^{2+} in portlandite crystals. *RSC Adv* 2014; 4:29305–29309.
- [16] Khachani M, El Hamidi A, Halim M, Arsalane S. Non-isothermal kinetic and thermodynamic studies of the dehydroxylation process of synthetic calcium hydroxide $\text{Ca}(\text{OH})_2$. *J Mater Environ Sci* 2014; 5:615–624.
- [17] Milagres JL, Bellato CR, Vieira RS, Ferreira SO, Reis C. Preparation and evaluation of the Ca-Al layered double hydroxide for removal of copper(II), nickel(II), zinc(II), chromium(VI) and phosphate from aqueous solutions. *J Environ Chem Eng* 2017; 5:5469–5480.
- [18] Schroeder PA. Infrared Spectroscopy in Clay Science Schroeder, P.A. (2002) Infrared Spectroscopy in clay science: In CMS Workshop Lectures, Vol. 11, Teaching Clay Science, A. Rule and S. Guggenheim, eds., The Clay Mineral Society, Aurora, CO, 181-206. Teaching clay science. Aurora, CO: Clay Minerals Society; 2002:182–206.
- [19] Coenen K, Gallucci F, Mezari B, Hensen E, van Sint Annaland M. An in-situ IR study on the adsorption of CO_2 and H_2O on hydrotalcites. *J CO_2 Util* 2018; 24:228–239.
- [20] Vieille L, Rousselot I, Leroux F, Besse J-P, Taviot-Guého C. Hydrocalumite and Its Polymer Derivatives. 1. Reversible Thermal Behavior of Friedel's Salt: A Direct Observation by Means of High-Temperature in Situ Powder X-ray Diffraction. *Chem Mater* 2003; 15:4361–4368.
- [21] Morales-Flórez V, Santos A, Romero-Hermida I, Esquivias L. Hydration and carbonation reactions of calcium oxide by weathering: Kinetics and changes in the nanostructure. *Chem Eng J* 2015; 265:194–200.